

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$f(x) = 9(4)^x$

The function f is defined by the given equation. If $g(x) = f(x + 2)$, which of the following equations defines the function g ?

Answer

- A. $g(x) = 18(4)^x$
- B. $g(x) = 144(4)^x$
- C. $g(x) = 18(8)^x$
- D. $g(x) = 81(16)^x$

Correct Answer: B

Rationale

Choice B is correct. It's given that $f(x) = 9(4)^x$ and $g(x) = f(x + 2)$. Substituting $x + 2$ for x in $f(x) = 9(4)^x$ gives $f(x + 2) = 9(4)^{x+2}$. Rewriting this equation using properties of exponents gives $f(x + 2) = 9(4)^x(4)^2$, which is equivalent to $f(x + 2) = 9(4)^x(16)$. Multiplying 9 and 16 in this equation gives $f(x + 2) = 144(4)^x$. Since $g(x) = f(x + 2)$, $g(x) = 144(4)^x$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.